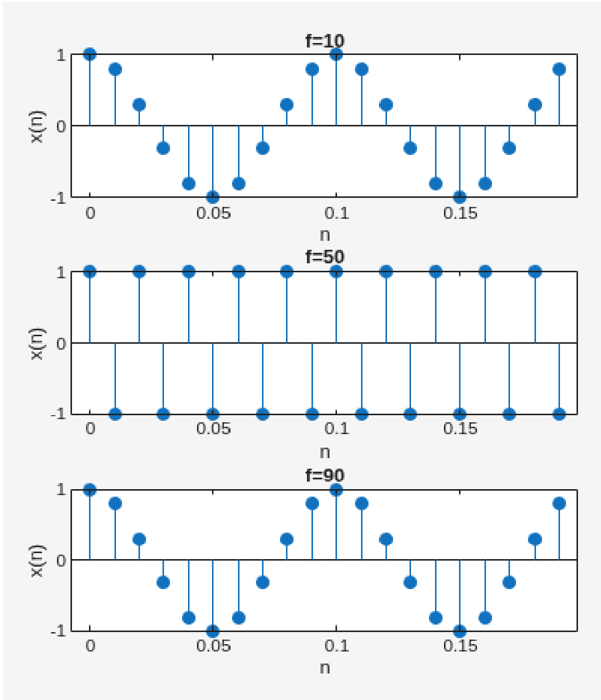
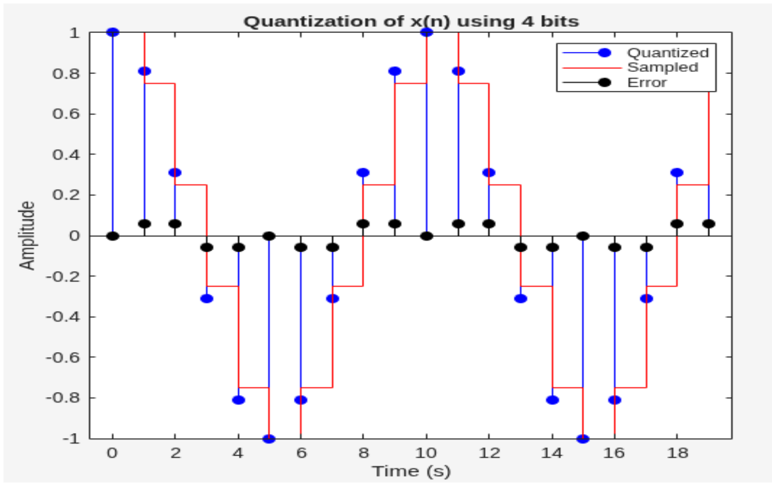
Task 1

```
fs = 100;
f1 = 10; f2 = 50; f3 = 90;
ts = 1/fs;
n = 0:19;
t = 0:ts:(19*ts);
x1 = cos(2*pi*f1*t);
x2 = cos(2*pi*f2*t);
x3 = cos(2*pi*f3*t);
subplot(3,1,1)
stem(t,x1,'filled');
title('f=10')
xlabel('n'); ylabel('x(n)');
subplot(3,1,2)
stem(t,x2,'filled');
title('f=50')
xlabel('n'); ylabel('x(n)');
subplot(3,1,3)
stem(t,x3,'filled');
title('f=90')
xlabel('n'); ylabel('x(n)');
```



Task 2

```
m = 4; f = 10;
l = 2.^m; fs = 100;
ts = 1/fs; n = 0:19;
t = n*ts;
x = cos(2*pi*f*t);
D = (max(x)-min(x))/l;
xq = quant(x,D);
error = x - xq;
figure;
stem(n, x, 'b','filled');
title('x(n) for f = 10 Hz');
hold on ; stairs(n, xq , ‘r’) ;
title('x(n) for f = 10 Hz');
stem(n, error, 'k','filled');
title('Quantization of x(n) using 4 bits');
xlabel('Time (s)') ylabel('Amplitude')
legend('Quantized','Sampled','Error')
```

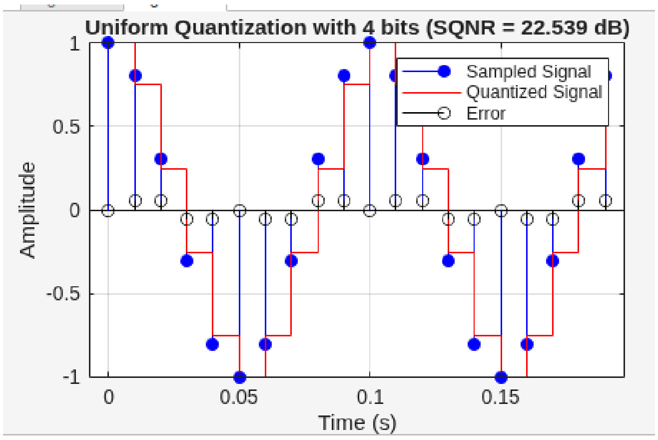


Task 3

```
f = 100;
m1 = 4; m2 = 5; m3 = 6 ;
L1 = 2.^m1; L2 = 2.^m2; L3 = 2.^m3 ;
fs = 200; ts = 1/fs; n=0:19;
% sampaled signal
t=n*ts; x=cos(2*pi*f*t);
D1 = (max(x)-min(x))/L1;
D2 = (max(x)-min(x))/L2;
D3 = (max(x)-min(x))/L3;
quantized signal
xq1 = quant(x,D1);
xq2 = quant(x,D2);
xq3 = quant(x,D3);
subplot(3,1,1) stairs(n,xq1)
title('4 bits')
subplot(3,1,2) stairs(n,xq2)
title('5 bits')
subplot(3,1,3) stairs(n,xq3)
title('6 bits')
err1 = x-xq1; err2 = x-xq2; err3 = x-xq3;
ratio1 = snr(x, err1)
ratio2 = snr(x, err2)
ratio3 = snr(x, err3)
fprintf('SQNR results -> 4 bits: %.2f dB, 5 bits: %.2f dB, 6 bits: %.2f dB\n', ratio1, ratio2, ratio3);
```

```
ratio1 =

   23.6668


ratio2 =

   34.6986


ratio3 =

   40.1123

SQNR results -> 4 bits: 23.67 dB, 5 bits: 34.70 dB, 6 bits: 40.11 dB
>>
```

```
function [xq, error, SQNR] = quantizer(x, m, t)
% Number of quantization levels
L = 2^m;
% Step size
D = (max(x) - min(x)) / L;
% Quantization
xq = quant(x, D);
% Error signal
error = x - xq;
% Compute SQNR
SQNR = snr(x, error);
% Plotting
figure;
stem(t, x, 'b', 'filled'); hold on; % Sampled
stairs(t, xq, 'r'); % Quantized
stem(t, error,'k'); % Error
legend('Sampled Signal', 'Quantized Signal', 'Error');
xlabel('Time (s)'); ylabel('Amplitude');
title(sprintf('Uniform Quantization with %d bits (SQNR = %.3f dB)', m, SQNR));
grid on;
end
```



<u>Checking:</u>

m = 4; l = 2^m;
f = 10; fs = 100; ts = 1/fs;
n = 0:19 ; t = n * ts;
x = cos(2 * pi * f * t);
[xq, error, SQNR] = quantizer(x, m, t);